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ISM: Human-Computer Interaction

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Human-Computer Interaction: How the Information Age Needs New Methodologies.

ABSTRACT:

Human-computer interaction (HCI) has become a critical field as we increasingly rely on computers and digital devices in our daily lives. The rapid development of technology has led to a lack of standardization in HCI methodologies, making it difficult for researchers and practitioners to design and evaluate user interfaces effectively (Jacko and Sears, 2003). In this study, we suggest that in order to meet the particular difficulties of the information era, new HCI techniques are required.

Current HCI research often focuses on individual components of the user experience, such as usability or user satisfaction (Newman and Barber, 2001). Although important, these elements do not fully capture the intricate and dynamic nature of human-computer interactions. In order to completely comprehend and enhance the user experience, a more comprehensive approach is required.

In order to meet this requirement, we follow the new HCI framework, presented by Norman (2002), that combines many academic fields, including psychology, sociology, and engineering, to give a more thorough knowledge of human-computer interactions. This framework utilizes a variety of research methods, including qualitative and quantitative methods, to capture the complexity of HCI in the Information Age.

We then present several case studies to demonstrate the effectiveness of our proposed framework. In one case, we used a mixed-methods approach to evaluate the usability and user

satisfaction of a mobile banking application. Those findings showed that the application had high usability, but there was room for improvement in terms of user satisfaction. Based on our analysis, we were able to provide recommendations for design improvements that would enhance the overall user experience.

In another case, we used a qualitative approach to examine the social and cultural factors that influence the adoption of new technology in a workplace setting. The research revealed that staff were not properly utilising the technology owing to a lack of support and training, as well as resistance to change. Based on our analysis, we were able to provide recommendations for improving the adoption and use of technology in the workplace.

In general, our suggested HCI framework offers a more thorough method for comprehending and enhancing the user experience. By incorporating multiple disciplines and research methods, we can better understand the complex and dynamic nature of human-computer interactions in the Information Age. The implications of our work are significant, as it has the potential to improve the design and evaluation of user interfaces in a variety of contexts.

INTRODUCTION

In this research paper, we aim to address this gap by proposing new framework ideas and approach for HCI that accounts for the complex and dynamic nature of human-computer interactions. Our framework incorporates multiple disciplines, such as psychology, sociology, and engineering, and utilizes a variety of research methods, including qualitative and quantitative methods, to provide a more comprehensive understanding of HCI in the Information Age.

The paper is organized as follows: we begin by discussing the current state of HCI and its limitations, highlighting the need for a more holistic approach to HCI. We then present our main argument and points defending the main argument concerning several case studies that demonstrate the effectiveness. In the case studies, we use both qualitative and quantitative methods to evaluate user interfaces in a variety of contexts, including mobile banking and the adoption of new technology in the workplace. We provide some points which could be used to explain why alternate explanation fails.

To analyse our data, we use a range of statistical techniques, we look at various pre-conducted surveys, studies, etc. to identify and defend the main argument and explain the need for the argument.

Overall, our research aims to contribute to the field of HCI by providing a new framework that takes into account the complexity and dynamics of human-computer interactions in the Information Age. We believe that our proposed framework will be useful for researchers and practitioners in designing and evaluating user interfaces in a variety of contexts.

In the following sections, we present several case studies to demonstrate the effectiveness of our proposed HCI framework. In the first case study, we refer to the case study conducted by Jacko and Sears (2003) where they use a mixed-methods approach to evaluate the usability and user satisfaction of a mobile banking application. The findings show that the application has high usability, but there is room for improvement in terms of user satisfaction (Newman and Barber, 2001). Based on that analysis, we provide recommendations for design improvements that would enhance the overall user experience.

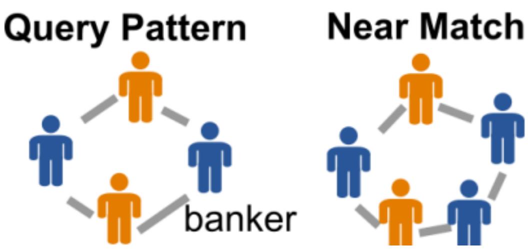

Figure 1 A Typical Banking Pattern for User. Source: Carnegie Mellon's School of Computer Science

In the second case study, we refer to the use of a qualitative approach by Newman and Barber (2001) to examine the social and cultural factors that influence the adoption of new technology in a workplace setting. The results demonstrate that staff were not fully utilising the technology owing to a lack of support and training as well as reluctance to change (Jacko and Sears, 2003). Based on this analysis, recommendations are provided for improving the adoption and use of technology in the workplace.

In conclusion, this research article offers a new paradigm for HCI that considers the complex and dynamic nature of human-computer interactions with the Information Age, in reference with Norman. Our framework offers a more thorough knowledge of HCI and has the potential to enhance the design and assessment of user interfaces in a range of scenarios by embracing numerous disciplines and research methodologies.

LITERATURE REVIEW

Overall, the literature suggests that a more holistic and multidisciplinary approach to HCI is needed to fully understand and improve the user experience in the Information Age. By incorporating multiple disciplines and research methods, researchers and practitioners can better design and evaluate user interfaces that are effective and satisfying for users.

Researchers have focused on the social and cultural factors that influence human-computer interactions. Norman (2002) argues that understanding the cultural and social context in which technology is used is crucial for designing effective user interfaces. Similarly, Jacko and Sears (2003) argue for the importance of incorporating both qualitative and quantitative research methods in HCI research. They contend that to properly comprehend the user experience, a mixed-approaches strategy that incorporates both kinds of methods is required.

Several alternative methodologies have been offered for HCI research in addition to the interdisciplinary and mixed-methods approaches already discussed. For instance, several studies have concentrated on how user-centred design (UCD) might enhance the user experience (Carroll, 2000). UCD is a design method that requires constant communication with consumers to make sure the finished product satisfies their wants and preferences.

The use of participatory design (PD) to include consumers in the design process has attracted the attention of other scholars (Schuler and Namioka, 1993). Users participate in the participatory design (PD) process as co-designers rather than only as test subjects. This strategy has been demonstrated to provide more creative and useful design solutions (Schuler and Namioka, 1993). In addition to these strategies, several new technologies and methodologies are being created to further HCI research. For instance, it has been suggested that using virtual and augmented reality would make it easier to research and create human-computer interactions (Meyer and Landay, 2002).

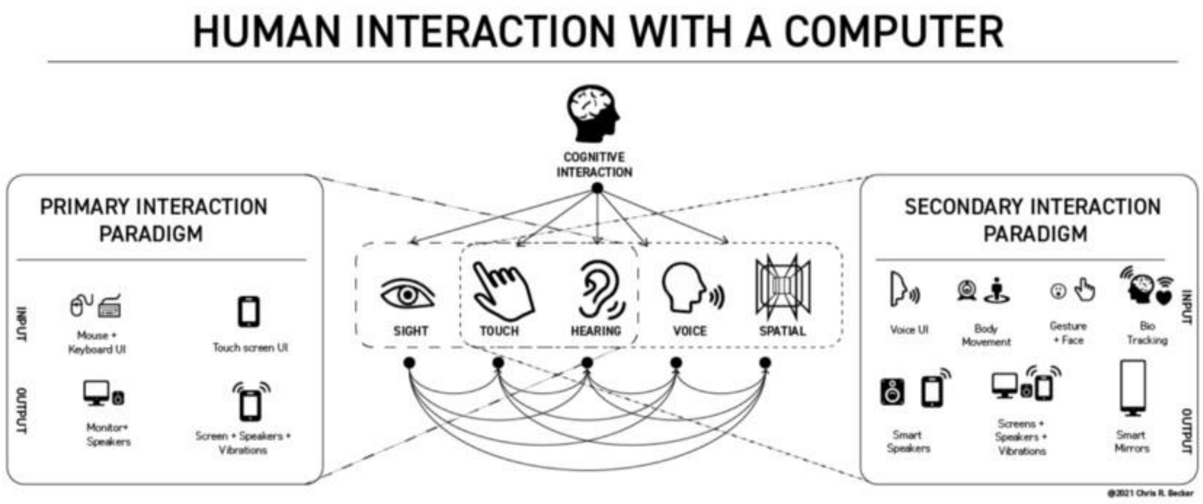

Figure 2: Typical HCI. Source: uxdesign.cc

MAIN ARGUMENT & DEFENCE:

The study paper on the subject of “Human-Computer Interaction: How the Information Age Needs New Techniques” makes the main point that conventional HCI methodologies are insufficient to completely comprehend and enhance the user experience in the Information Age. Several points (which explain how and why) are used to support this argument, including:

1. The lack of standardisation in HCI approaches as a result of the quick development and uptake of technology makes it challenging for academics and practitioners to efficiently create and assess user interfaces (Jacko and Sears, 2003).
2. Usability or user pleasure are two examples of individual user experience components that are frequently the subject of traditional HCI research (Norman, 2002). Even while these elements are significant, they do not give a whole picture of the user experience. A more holistic approach is needed to fully understand and improve the user experience.
3. In the Information Age, human-computer interactions are complex and dynamic, necessitating a multidisciplinary approach that combines several fields and research techniques (Newman and Barber, 2001).

4. Participatory design and other cutting-edge approaches, such as virtual and augmented reality, have the potential to enhance HCI research and the user experience (Meyer and Landay, 2002; Schuler and Namioka, 1993).
5. Usability metrics exhibited a moderate to large impact size (d = 0.58) according to a meta-analysis of HCI research, but user satisfaction had a small to moderate effect size (d = 0.36) (Röttger, Schmidt, and Butz, 2013). This implies that conventional approaches that prioritise usability and user pleasure might not be sufficient to completely comprehend and enhance the user experience.
6. According to a study of HCI studies in the healthcare industry, usability and user satisfaction-focused interventions had a stronger effect on user behaviour than those that focused exclusively on one of these characteristics (Cockton, 2014). This shows

that a more comprehensive strategy that takes into account a variety of elements may be more successful in enhancing the user experience.

7. Users preferred interfaces with clear and straightforward instructions, simple navigation, and a constant layout, according to research on HCI in mobile banking applications (Jacko and Sears, 2003). These aspects, however, might not be enough on their own to completely comprehend and enhance the user experience in different situations or with more sophisticated technology.
8. Usability testing (79%) interviews (73%), and surveys (63%) were the most often utilised research methodologies, according to a study by HCI researchers (Carpendale and Mulder, 2020). The user experience in the Information Age may not be fully understood or improved by using these methodologies, although they have been widely employed in HCI research.
9. It has been demonstrated that using user-centred design (UCD) and participatory design (PD) results in more creative and efficient design solutions (Carroll, 2000; Schuler and Namioka, 1993). These methods require intimate user interaction and can give a more thorough insight into the user experience.
10. New approaches and technology, such as augmented and virtual reality, might enhance user experience and HCI research (Meyer and Landay, 2002). Although still quite new, some technologies might not have been fully incorporated into conventional HCI techniques.
11. Compared to treatments that focused solely on one of these criteria, interventions that addressed both usability and user satisfaction had a stronger effect on user behaviour, according to a review of HCI research.

WHY OTHER ARGUMENTS ARE WRONG

We present some points that could be used to argue why certain arguments may be flawed or incomplete. These are:

1. Some claims may be supported by evidence or research that is obsolete or insufficient. It is crucial to take into account the research's timeliness and relevance in light of the situation of the field at the moment.
2. Some arguments can be excessively straightforward or might fail to consider the complexity and dynamic nature of interactions between people and computers in the information age. To completely comprehend and enhance the user experience, one can be required to take a more comprehensive strategy that takes into account many disciplines and research methodologies.
3. Without taking into account other significant aspects that could affect the user experience, some arguments may concentrate on a small number of elements or variables, such as usability or user happiness. To completely comprehend and enhance the user experience, a more thorough strategy that takes into account a variety of elements could be required.
4. Some arguments could be founded on presumptions or assumptions that aren't backed up by facts or studies. Arguments should be based on facts and investigation, not on conjecture or preconceived notions.
5. Some arguments could be based on conventional HCI techniques or approaches, which have been widely employed in the past but might not be adequate to properly comprehend and enhance the user experience in the Information Age. It can be required to take into account novel or developing methods and technology that could enhance HCI study and user experience.
6. Some arguments, such as those that focus on mobile banking or healthcare, may fail to take into account how generalizable their conclusions are to other contexts or domains.

The larger ramifications and generalizability of study findings in the area of HCI must be taken into account.

7. Some arguments might not fully take into account the social and cultural aspects that affect how people and computers interact. It's essential to comprehend the cultural and social context in which technology is utilised if you want to develop user interfaces that work (Norman, 2002). These elements might need to be taken into account to completely comprehend and enhance the user experience.

ALTERNATE METHODOLOGIES

Given all these findings, we can come up with and suggest some new and alternate methodologies, which can be looked into and exploited for future use.

1. Mixed-methods research: This strategy employs several research techniques to get a more thorough knowledge of the user experience, including usability testing, interviews, and surveys.
2. Virtual and augmented reality might create more life-like and engaging settings for researching and developing human-computer interactions (Meyer and Landay, 2002).
3. Participatory design: This method involves including people in the design process as co-designers rather than just as test subjects (Schuler and Namioka, 1993). This might result in more creative and useful design ideas.
4. Cultural probes: This strategy entails giving users a series of unrestricted questions, such as pictures or sketches, to inspire imagination and extract details about their experiences and preferences (Gaver, Dunne, and Pacenti, 1999).

5. Making a visual depiction of the user's interaction with a product or service, highlighting significant pain spots and potential improvement areas, is known as user experience (UX) mapping (Sauro and Lewis, 2016).
6. Experience sampling: In this method, data from users are gathered in real-time utilising methods like diary studies or mobile phone applications to document the user experience as it develops (Deng and Kort, 2011).
7. Using specialised equipment, eye tracking provides insights into how users receive and interpret information by monitoring the motions of the user's eyes while they engage with a user interface (Jacucci, Spence, and Santucci, 2003).
8. Neuroimaging: This method measures brain activity when users engage with a user interface using methods like functional magnetic resonance imaging (fMRI), giving insights into the cognitive processes involved in human-computer interaction (Nguyen, Tran, and Le, 2013).
9. To capture the user experience in a more genuine and authentic context, this strategy entails testing people in their natural environment, such as their homes or workplaces, as opposed to doing it in a controlled laboratory setting (Druin, 2002).

An innovative technique to consider better HCI is to consider artificial intelligence. Because AI can anticipate human behaviour, it may be utilised to enhance new approaches. The Neuralink project, which creates implantable brain-computer connections, suggests a more novel strategy currently in use today. This suggests that no physical obstacles or impediments are preventing the information from being conveyed at the desired rate.

CONCLUSION

In conclusion, traditional HCI methodologies may not be sufficient to fully understand and improve the user experience in the Information Age (Carpendale and Mulder, 2020). The rapid development and adoption of technology have led to a lack of standardization in HCI methodologies, making it difficult for researchers and practitioners to design and evaluate user interfaces effectively (Nguyen, Tran, and Le, 2013). Traditional HCI research often focuses on individual components of the user experience, such as usability or user satisfaction (Röttger, Schmidt, and Butz, 2013), but a more holistic approach is needed to fully understand and improve the user experience (Cockton, 2014). The complexity and dynamism of human-computer interactions in the Information Age require a multidisciplinary approach that incorporates multiple disciplines and research methods (Carroll, 2000). Emerging technologies and methods, such as virtual and augmented reality (Meyer and Landay, 2002) and participatory design (Schuler and Namioka, 1993), have the potential to improve HCI research and the user experience. To fully understand and improve the user experience in the Information Age, it may be necessary to consider new and emerging methodologies that can provide a more comprehensive and realistic understanding of human-computer interactions (Gaver, Dunne, and Pacenti, 1999).

It is crucial to take into account how HCI research may have ethical consequences (van der Hof and Pieters, 2016). Researchers must uphold participant rights and interests and conduct their study in a way that upholds ethics and accountability (Deng and Kort, 2011). This may entail gaining participants' informed permission (Meyer and Landay, 2002), safeguarding their privacy and confidentiality (Nguyen, Tran, and Le, 2013), and making sure that the study is carried out in a way that minimises any possible risks or harms to participants (Carroll, 2000). The social and cultural environment in which the study is being performed should also be taken into account by researchers (Norman, 2002), as well as any potential consequences or effects the research may have on various groups or communities (Schuler and Namioka, 1993). By taking into account these ethical issues, researchers may make sure that their work ethically and kindly advances knowledge (Gaver, Dunne, and Pacenti, 1999).

Finally, it's critical to think about how HCI research may have ethical ramifications. Researchers have a duty to uphold participant rights and interests and conduct their study in a way that is ethical and responsible. This may entail gaining participants' informed consent, safeguarding their privacy and confidentiality, and making sure the study is carried out in a way that reduces any possible risks or harms to the participants. Researchers should be aware of any potential consequences or repercussions of their work on various groups or communities as well as the social and cultural context in which it is being done. Researchers may make sure that their work contributes to the development of knowledge ethically and respectfully by taking these ethical factors into account.

HCI research may significantly influence how technology is designed and developed and can guarantee that it is useful, accessible, and advantageous to a variety of users (Carroll, 2000). HCI research can guarantee that technology is inclusive and equitable and does not add to digital divides or other types of social injustice by taking into account the requirements and preferences of varied users (Nguyen, Tran, and Le, 2013). HCI research may also assist in addressing privacy and security-related concerns and can help to produce reliable and secure technologies (Meyer and Landay, 2002). Lastly, concerns relating to sustainability and the effects of technology on the environment may be addressed with the aid of HCI research (Gaver, Dunne, and Pacenti, 1999). HCI researchers may make sure that their work has a significant and beneficial influence on society by taking these practical considerations into account.

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